

Role and description of ITIL, ISO certification, and laws in IT

1. What is ITIL?

Definition:

ITIL (Information Technology Infrastructure Library) is a **set of best practices** for **managing IT services** and aligning them with business needs.

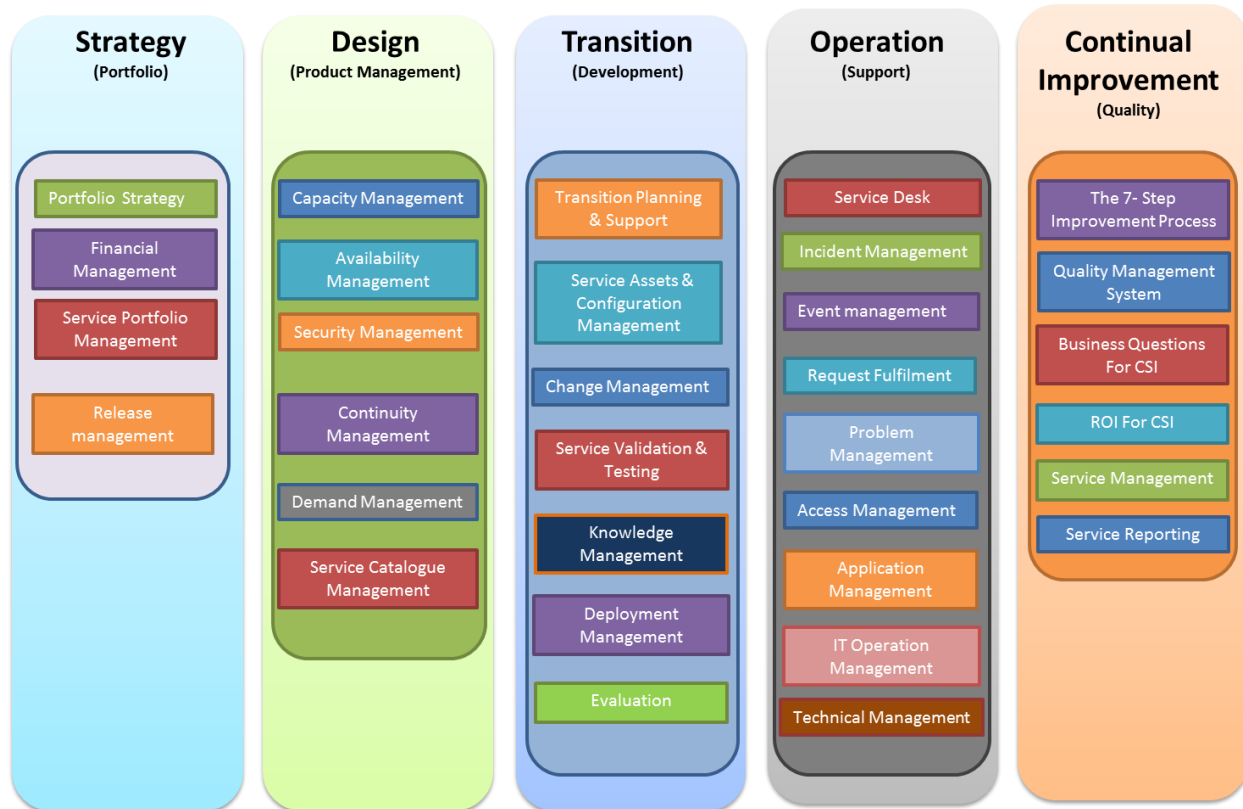
- Framework for IT service management (ITSM)

Goals of ITIL:

- Improve customer satisfaction
- Provide consistent service quality
- Reduce costs and risks
- Ensure stable, reliable and well-governed IT services

ITIL v4 structure:

1. **Service Value System (SVS)** – the core model, represents how components of an organization work together to create value.
2. **4-dimensions model** – people, processes, technology, partners
3. **Service value chain** – planning, design, delivery, support
4. **Key processes:**
 - Incident management
 - Change control
 - Configuration management
 - Problem management



Strategy - The process identifies (market) opportunities that could be exploited by introducing new services. The most important chapters of the volume are Service Portfolio Management and Financial Management.

Design - As a result of the process, a project plan is prepared for the specific implementation of the service outlined by the strategy created in the previous step. The most important chapters of the volume are Operation and Maintenance, Capacity Planning, and IT and Operational Security.

Transition - Description of the processes required to establish the planned service and modify the environment. Important chapters are Change and Version Management, Configuration Management, and Documentation Management.

Operation - The volume, which is closely related to the previous one, discusses the processes and organizational issues necessary for the continuous and error-free operation of the service.

Continual Improvement - The volume discusses the conditions for providing the service with continuously improving quality. Featured chapters are the chapters on Service Level Measurement, Reporting and Management.

ITIL is not a law or a regulation – it's a **framework**, often adopted by IT departments to improve service delivery.

Example: A company uses ITIL's Incident Management practice to reduce downtime by responding faster to IT problems.

2. What is **ISO certification**?

Definition:

ISO (International Organization for Standardization) issues global **standards** to ensure the **quality, security, and efficiency** of systems and organizations.

Common ISO standards in IT:

Standard	Description
ISO/IEC 27001	Standard for Information Security Management Systems (ISMS)
ISO/IEC 20000	Standard for IT Service Management , aligned with ITIL
ISO 9001	Standard for Quality Management Systems

Why ISO certifications matter?

- Show clients and partners the organization meets international standards
- Help reduce risks and improve processes
- Often required for working with government or large clients

How certification works?

1. Organization prepares and documents policies
 2. An external auditor inspects compliance
 3. Certificate is issued for a fixed period (typically 3 years)
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Laws in IT

1. Importance of laws in IT

Why laws are needed?

- Protect data privacy
- Prevent cybercrime
- Establish responsibility and accountability
- Regulate digital transactions, e-commerce, and communication

IT professionals must be aware of the legal environment they work in, especially regarding **user data** and **network security**.

2. Key IT-related laws and regulations

General Data Protection Regulation (GDPR):

- Protects **personal data of individuals**
- Gives people rights over how their data is collected, stored, and used
- Organizations must have **user consent** and ensure **data security**
- Heavy fines for violations

Cybercrime laws:

- Cover illegal access, data theft, malware creation, denial-of-service attacks
- Define **cybercrime** and **punishments**

Copyright and intellectual property:

- Software is legally protected
- Illegal copying, downloading, or modifying software is punishable
- Open-source licenses (GPL ((GNU General Public License)), MIT) still come with conditions - MIT is an extremely permissive license that allows free use, modification

and sale with attribution, while GPL is a "copyleft" license that requires all derivatives to remain open under the same license, ensuring code sharing.

Electronic signature laws:

- Give digital signatures legal value
- Enable legally binding **electronic contracts**

Key terms summary

Term	Definition
ITIL	Framework for managing and improving IT service delivery
ISO/IEC 27001	International standard for managing information security
ISO/IEC 20000	Standard for IT service management
GDPR	EU law protecting personal data and privacy
Cybercrime	Illegal activity involving computers or networks
Digital signature	Encrypted signature validating the authenticity of electronic documents
ISMS	Information Security Management System – part of ISO 27001